

PREFACE FOR FACULTY

This introductory lecture establishes the physical, mathematical, and architectural foundations of Digital Signal Processing (DSP). In undergraduate Electrical and Electronics Engineering (EEE), students transition from continuous-time physical systems (analog voltages, currents, electromagnetic waves) to sampled, quantized, discrete-time representations processed by digital microprocessors, DSP chips, and FPGAs.

Pedagogical Strategy:

1. Contrast continuous-time physical signals $x_a(t)$ with discrete-time sequences $x[n]$. Emphasize that n is an integer index, not physical time.
 2. Formulate the fundamental sampling relationship $t = nT_s = n/F_s$.
 3. Classify discrete-time signals mathematically and physically: Energy signals vs. Power signals, Periodic vs. Aperiodic sequences, and Even vs. Odd symmetric components.
 4. Define and visualize elementary discrete sequences: Unit impulse $\delta[n]$, Unit step $u[n]$, Unit ramp $r[n]$, Real exponential $a^n u[n]$, and Complex exponential $e^{j\omega_0 n}$.
 5. Highlight the fundamental distinction between continuous and discrete sinusoids: Discrete sinusoidal frequencies are unique only over an interval of length 2π (e.g., $[-\pi, \pi]$ or $[0, 2\pi]$).
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Differentiate** between continuous-time, discrete-time, and digital signals across time-domain and amplitude-domain representations.
 2. **Compute** the fundamental period N_0 of discrete-time sinusoids and verify periodicity conditions.
 3. **Decompose** an arbitrary discrete-time sequence into its even symmetric $x_e[n]$ and odd symmetric $x_o[n]$ components.
 4. **Evaluate** the total signal energy E and average power P of elementary and composite discrete-time sequences.
 5. **Formulate** elementary signal operations including time-shifting, time-reversal, amplitude scaling, and downsampling/upsampling.
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2. MATHEMATICAL FOUNDATIONS & DEFINITIONS

2.1 Continuous to Discrete Sampling

A discrete-time signal $x[n]$ is obtained by sampling a continuous-time signal $x_a(t)$ at uniform intervals of T_s seconds:

$$x[n] = x_a(nT_s) = x_a(t)|_{t=nT_s}, \quad n \in \mathbb{Z}$$

Where:

- $T_s = 1/F_s$ is the sampling period in seconds.
- F_s is the sampling frequency in Hertz (Hz or samples/sec).
- Ω denotes continuous-time angular frequency (rad/s), and $\omega = \Omega T_s = \Omega/F_s$ denotes normalized discrete-time angular frequency (rad/sample).

2.2 Elementary Discrete-Time Sequences

1. Unit Impulse (Kronecker Delta):

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$\text{Sifting Property: } \sum_{n=-\infty}^{\infty} x[n] \delta[n - n_0] = x[n_0].$$

2. Unit Step Sequence:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = \sum_{k=0}^{\infty} \delta[n - k] = \sum_{m=-\infty}^n \delta[m]$$

$$\text{Relation to impulse: } \delta[n] = u[n] - u[n - 1].$$

3. Unit Ramp Sequence:

$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

4. Real Exponential Sequence:

$$x[n] = a^n u[n] = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- For $0 < a < 1$: decaying monotonic sequence.
- For $-1 < a < 0$: alternating decaying sequence.
- For $|a| > 1$: exponentially growing sequence (unstable).

5. Complex Exponential and Sinusoidal Sequences:

$$x[n] = A e^{j(\omega_0 n + \phi)} = A \cos(\omega_0 n + \phi) + jA \sin(\omega_0 n + \phi)$$

2.3 Periodicity Condition in Discrete Time

A discrete-time sequence $x[n]$ is periodic with fundamental period $N \in \mathbb{Z}^+$ if and only if $x[n + N] = x[n]$ for all n . For a sinusoid $x[n] = \cos(\omega_0 n)$:

$$\cos(\omega_0(n + N)) = \cos(\omega_0 n + \omega_0 N) = \cos(\omega_0 n) \iff \omega_0 N = 2\pi k, \quad k \in \mathbb{Z}^+$$

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q} \quad (\text{Ratio of frequency to } 2\pi \text{ must be a rational number})$$

The fundamental period is $N_0 = \frac{2\pi k}{\omega_0}$, where k is the smallest positive integer such that N_0 is an integer.

2.4 Energy and Power Classification

• Total Signal Energy E :

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

• Average Signal Power P :

$$P = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

For a periodic sequence with fundamental period N_0 :

$$P = \frac{1}{N_0} \sum_{n=0}^{N_0-1} |x[n]|^2$$

• **Classification:**

1. *Energy Signal:* $0 < E < \infty \implies P = 0$.
2. *Power Signal:* $0 < P < \infty \implies E = \infty$.
3. *Neither:* $E = \infty$ and $P = \infty$ or $P = 0$.

2.5 Even and Odd Symmetry Decomposition

Any arbitrary sequence $x[n]$ can be uniquely decomposed into even and odd components:

$$x[n] = x_e[n] + x_o[n]$$

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]) \quad (\text{Even Component: } x_e[-n] = x_e[n])$$

$$x_o[n] = \frac{1}{2}(x[n] - x[-n]) \quad (\text{Odd Component: } x_o[-n] = -x_o[n], ; x_o[0] = 0)$$

3. WORKED NUMERICAL EXAMPLES

Example 1.1: Periodicity Analysis of Composite Discrete Sinusoids

Problem: Determine if the following signals are periodic. If periodic, find the fundamental period N_0 : (a)

$$x_1[n] = \cos\left(\frac{3\pi}{7}n + \frac{\pi}{4}\right) \quad (b) \quad x_2[n] = \cos(0.8n) \quad (c) \quad x_3[n] = \cos\left(\frac{\pi}{6}n\right) + \sin\left(\frac{\pi}{8}n\right)$$

Solution: (a) $\omega_1 = \frac{3\pi}{7}$.

$$\frac{\omega_1}{2\pi} = \frac{3\pi/7}{2\pi} = \frac{3}{14} \in \mathbb{Q}$$

Since the ratio is rational, $x_1[n]$ is periodic. Fundamental period: $N_1 = \frac{2\pi \cdot k}{\omega_1} = \frac{2\pi \cdot 3}{3\pi/7} = 14$ samples (with $k = 3$).

(b) $\omega_2 = 0.8 = \frac{4}{5}$.

$$\frac{\omega_2}{2\pi} = \frac{0.8}{2\pi} = \frac{0.4}{\pi} \notin \mathbb{Q}$$

Because π is irrational, $\frac{\omega_2}{2\pi}$ is irrational. Therefore, $x_2[n]$ is **aperiodic (non-periodic)**.

(c) For composite signal $x_3[n]$:

- For $\cos\left(\frac{\pi}{6}n\right)$: $\omega_a = \pi/6 \implies \frac{\omega_a}{2\pi} = \frac{1}{12} \implies N_a = 12$.
- For $\sin\left(\frac{\pi}{8}n\right)$: $\omega_b = \pi/8 \implies \frac{\omega_b}{2\pi} = \frac{1}{16} \implies N_b = 16$. Both terms are periodic. The overall fundamental period is the Least Common Multiple (LCM):

$$N_0 = \text{LCM}(N_a, N_b) = \text{LCM}(12, 16) = 48 \text{ samples}$$

Example 1.2: Energy and Power Calculation

Problem: Classify the following signals as Energy, Power, or Neither, and compute the corresponding E or P :

(a) $x[n] = (0.6)^n u[n]$ (b) $x[n] = e^{j(\frac{\pi}{4}n + \frac{\pi}{3})}$

Solution: (a) Total Energy:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=0}^{\infty} |(0.6)^n|^2 = \sum_{n=0}^{\infty} (0.36)^n$$

Using the infinite geometric series formula $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$:

$$E = \frac{1}{1-0.36} = \frac{1}{0.64} = \frac{25}{16} = 1.5625 \text{ Joules}$$

Since $0 < E < \infty$, $x[n]$ is an **Energy Signal** with average power $P = 0$.

(b) The signal has $|x[n]| = |e^{j(\frac{\pi}{4}n + \frac{\pi}{3})}| = 1$ for all n . Fundamental period: $\omega_0 = \pi/4 \implies N_0 = \frac{2\pi}{\pi/4} = 8$.

Average Power:

$$P = \frac{1}{N_0} \sum_{n=0}^{N_0-1} |x[n]|^2 = \frac{1}{8} \sum_{n=0}^7 1^2 = \frac{8}{8} = 1 \text{ Watt}$$

Total energy $E = \sum_{n=-\infty}^{\infty} 1 = \infty$. Therefore, $x[n]$ is a **Power Signal** with $P = 1 \text{ W}$ and $E = \infty$.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Define elementary discrete-time sequences: $\delta[n]$, $u[n]$, and $r[n]$. Express $u[n]$ in terms of $\delta[n]$ and vice-versa. (5 Marks) **(b)** A continuous-time signal $x_a(t) = 3 \cos(100\pi t) + 2 \sin(250\pi t)$ is sampled at $F_s = 200 \text{ Hz}$.

1. Find the discrete-time sequence $x[n]$. (3 Marks)
2. Determine whether $x[n]$ is periodic. If periodic, find the fundamental period N_0 . (4 Marks)
3. Calculate the average power P of $x[n]$. (3 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Definitions of $\delta[n]$, $u[n]$, $r[n]$ with equations and sketches (3 Marks)
 - $u[n] = \sum_{k=-\infty}^n \delta[k] = \sum_{m=0}^{\infty} \delta[n-m]$ and $\delta[n] = u[n] - u[n-1]$ (2 Marks)
- **Part (b.1):**
 - Substitute $t = n/F_s = n/200$:

$$x[n] = 3 \cos\left(100\pi \frac{n}{200}\right) + 2 \sin\left(250\pi \frac{n}{200}\right) = 3 \cos\left(\frac{\pi}{2}n\right) + 2 \sin\left(\frac{5\pi}{4}n\right)$$

Since $\frac{5\pi}{4} = 2\pi - \frac{3\pi}{4}$, the principal alias is:

$$x[n] = 3 \cos(0.5\pi n) - 2 \sin(0.75\pi n)$$

(3 Marks)

- **Part (b.2):**
 - $\omega_1 = 0.5\pi \implies \frac{\omega_1}{2\pi} = \frac{1}{4} \implies N_1 = 4$.
 - $\omega_2 = 0.75\pi \implies \frac{\omega_2}{2\pi} = \frac{3}{8} \implies N_2 = 8$.

- $N_0 = \text{LCM}(4, 8) = 8$ samples. (4 Marks)
- **Part (b.3):**
 - Average power:

$$P = \frac{3^2}{2} + \frac{2^2}{2} = 4.5 + 2.0 = 6.5 \text{ Watts}$$

(3 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Sampling verification
Fs = 200
n = np.arange(0, 16)
xn = 3 * np.cos(np.pi / 2 * n) + 2 * np.sin(5 * np.pi / 4 * n)

# Compute fundamental period power
N0 = 8
P = np.mean(xn[:N0]**2)
print(f"Discrete Signal Power P = {P:.4f} W (Theoretical = 6.5 W)")
```